

Toward Using Multi-Modal Machine Learning for User Behavior Prediction in Simulated Smart Home for Extended Reality

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ABSTRACT

In this work, we propose a multi-modal approach to manipulate smart home devices in a smart home environment simulated in virtual reality (VR). We determine the user's target device and the desired action by their utterance, spatial information (gestures, positions, etc.), or a combination of the two. Since the information contained in the user's utterance and the spatial information can be disjoint or complementary to each other, we process the two sources of information in parallel using our array of machine learning models. We use ensemble modeling to aggregate the results of these models and enhance the quality of our final prediction results. We present our preliminary architecture, models, and findings.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality;

1 INTRODUCTION

Smart Home Devices (SHD) are designed to achieve more functionalities than traditional household devices. However, user interactions with current SHDs in the market require redundant steps. For example, to turn on a microwave, a user must either activate a smart speaker (e.g. Google Home, Alexa, etc) and give a very precise vocal command addressing that specific device, or control the SHD through a mobile app and its 2D menu. In both cases, there is a discontinuity in the user interaction flow, forcing the user to temporarily detach from their objective and perform extraneous actions.

In prior works, Köse et al. only utilize the user's body interaction to predict the user's future behaviour [2]; Xu et al. consider both user's body information and speech instructions in a virtual smart home environment but do so separately [4, 5]; existing works which explored multi-modal user behavior prediction use natural language processing (NLP) and computer vision techniques as no VR-related environment information could be used [1, 3]. However, we believe that modeling user behavior in VR scenes is important as for future metaverse applications. Under the VR setting, a user's utterance, body information (position, head gaze, hand direction, etc.), combined with SHD position and rotation and other properties could be taken together as a multi-modal approach to improve the SHD interaction and create a more continuous, natural and immersive experience. For example, instead of saying "Hey Alexa, turn on my smart microwave 01.", the user could simply say "Turn this on."

Our contributions are: 1) design a system in the simulated smart home scene (Fig. 1) where the user could interact with various SHDs, and 2) propose a multi-modal approach to predict user behaviors and verify the feasibility of our method through user study.

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Figure 1: Simulated Smart Home Environment in Virtual Reality. Cyan head in the middle represents the user.

2 SMART HOME DEVICE INTERACTION

2.1 Types of Smart Devices

Stationary Exocentric Devices: These are what comes to mind when discussing SHDs. These include light fixtures, ovens, microwaves, etc. These devices are stationary when being operated and they are exocentric as they do not move when the users move.

Mobile Exocentric Devices: These include devices such as autonomous robotic vacuum cleaners or a grocery delivery robot. They are mobile and exocentric since they move on their own as part of their operations, but user may still interact with them (e.g. point to a location and instruct the selected robot to "clean here").

EGOCENTRIC DEVICES: These refer to devices that are equipped by and move alongside the user. These include smart phones, glasses, watches, or earphones. Although some of these devices are currently not smart devices and require hand-operated buttons, it's not hard to imagine a scenario where there is a smart handheld mixer where the user would say "Mix on low for 30 seconds."

2.2 Ways to Interact with a Device

By SHD Name: The user can choose a device by explicitly referring to the device's unique name, e.g. "open oven 01".

By SHD Type: The user can choose a device by explicitly referring to the device's type, e.g. "open the microwave". If there are multiple copies, additional information will be needed.

By Type and Spatial References: Additional spatial information regarding the intended SHD target could be provided to help identify, e.g. "open the cabinet to the left of the fridge".

By Gaze: The user could interact with a device via looking at¹ it and saying commands such as "open it".

By Hand Direction: The user could interact with a device via pointing² it and saying commands such as "start this".

¹In the absence of a VR headset with eye-tracking features, we approximate this with a simple head gaze based on the headset position and rotation.

²We use VR controller's position and rotation to simulate user pointing.

By Recent Process Completion: If a device has recently finished a task, the user could interact with it via “open the microwave which just finished heating” or simply “open”.

By Special Command Set: For a timer in SHD, the user could have a command such as “start the timer for 5 minutes” which includes start and stop intents.

By User Equipment: If there is no exocentric camera near the user and the user says “take a picture”, an appropriate user’s egocentric equipment will be triggered to work.

3 MULTI-MODAL SYSTEM AND EXPERIMENTS

3.1 Pipeline Overview

To achieve the multi-modal user behavior prediction with various smart home device interactions, we propose the pipeline as shown in Fig. 2, where the Unity front-end is utilized to collect raw data and the back-end represents different modules that work together to provide the final results. To prove the concept, we designed experiments based on Stationary Exocentric Devices and focusing on interactions via SHD Name, SHD Type, SHD Type and Spatial References, and Gaze. We believe our method could be easily extended to more complex scenarios.

3.2 Front-End Data Collection with Unity

We built the simulated smart home environment in Unity as the platform to collect data. Under the instructions provided, participants were recruited to enter the scenario through VR devices and finish tasks such as opening and closing the microwave ten times while standing at different locations and saying commands in various ways. As a result, we collected about 27k records for intents to open and close smart home devices, including three cabinets, a dishwasher, a freezer, a microwave, and an oven. Each record contains the current SHD positions, the user’s head and hands position and rotation, and recognized user utterance³.

3.3 Back-End Modules

Preprocessing Spatial Algorithm We further process the raw environment and user information into two different distance lists regarding gaze information⁴. Cosine Distance List measures which devices are in the front of the user and how far they are regarding the orientation degree, i.e. is it in front of the user. Euclidean Distance List measures the spatial distance in the 3D space of the SHDs from the user.

NLP Module RASA framework, originally designed for conversational agents, is used to handle text data from Unity. Given a text utterance, the recognized entities and predicted intent will be returned and both of these results will serve as another source of input to the machine learning (ML) module, and the final predicted action is the most likely intent. As discussed in Sect. 2.2, if the user says “Open the stoves on the left.” when there are four gas stoves arranged in a square pattern, the keyword “left” will be recognized so that the left two stoves will get additional weight during prediction.

Multi-Modal Machine Learning Module Taking the pre-defined object availability list (for each object, what possible operations are), distance lists from Spatial Algorithm module, together with the entity list and predicted intent from NLP module, we train 3 models on random split collected data and further adapt the voting ensemble strategy to decide the final target. **Multiple linear regression** model is an extension of simple linear regression. With input lists, the model is trained to learn the weight for each list and used the best weight to get the most likely target. **Decision tree** model is used with an adopted CART algorithm to learn a series of if-else decision rules to predict the final target. For **Naive Bayes** classifier,

³The speech recognition service in Windows 10 is used in Unity

⁴The same process can be used on other user’s body-related information.

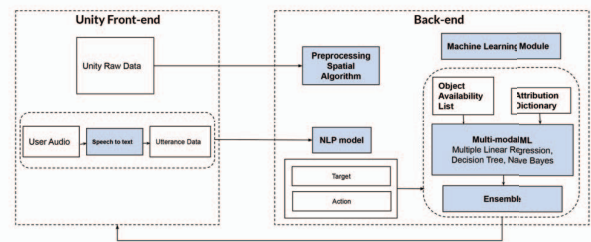


Figure 2: System Pipeline Overview

input lists are considered as different features to the classifier and the predicted result will be returned based on the calculated probability. As each model has one vote regarding the the target prediction, the **voting ensemble** strategy takes advantage of the majority votes to decide the final target.

3.4 User Study

We conducted experiments to evaluate the performance of the Multi-Modal Machine Learning Module under real user cases for seven targets (three cabinets, the dishwasher, the oven, the microwave, and the freezer). We instructed the participant to look at five different positions of the device to mitigate user biases and saying two commands: open and close. Out of 70 trials, the success rates for multiple linear regression, decision tree, and Naive Bayes are 85.71%, 94.29% and 97.14% respectively. Using the ensemble strategy, the success rate could reach 100.0% as expected due to the simplified setting of the current task.

4 CONCLUSION

To interact with smart home devices more naturally, we propose a multi-modal approach in this work, where the user’s intent is determined by the spoken command along with spatial information. We explore how various interaction cases could be handled by our system through NLP and ML techniques and verify the approach with a simple scenario. As an early work in this area, we hope it will serve as the basis for future works that can handle more complex user interaction cases.

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